

Manipal School of Information Sciences

Manipal Academy of Higher Education, Manipal

Outcome Based Education (OBE) Framework

Two Year full time Postgraduate Program

Course File

Course Name : **Modern Databases for Big Data**

Course Code : **BDA 5202**

Academic Year : **2024-25**

Semester : **II**

Name of the Course Coordinator : **Devi Prasad M**

Name of the Program Coordinator : **Dr. Prathviraj N**

Signature of Program Coordinator
with Date

Signature of Course Coordinator
with Date

Contents

Program Outcomes (POs).....	4
1. Course Details.....	5
1.1. Primary Information.....	5
1.2. Course Outcomes.....	6
1.3. Course Content (Syllabus).....	6
1.4. References	7
Topic Learning Outcomes	8
Teaching-Learning Practice (TLP) - Blackboard, PPT, Quizzes, and Solving Problems in the Classroom.....	8

Program Outcomes (POs)

By the end of the postgraduate program in ME (Big Data Analytics), graduates will be able to:

PO1	Independently carry out research/investigation and development work to solve practical problems.
PO2	Write and present a substantial technical report/document.
PO3	Demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
PO4	Develop and implement big data analysis strategies based on theoretical principles, ethical considerations, and detailed knowledge of the underlying data.
PO5	Demonstrate knowledge of the underlying principles and evaluation methods for analyzing data for decision-making.

1. Course Details

1.1. Primary Information

Course Name	:	Cloud Security
L-T-P-C	:	3-0-0-3
Contact Hours	:	36 Hours
Pre-requisite	:	Critical Thinking

1.2. Course Outcomes

CO No.	Course Outcomes (COs)	No. of Contact Hours	Bloom's Taxonomy Level	PO Attainment
CO1	Clearly articulate the underlying principles of the four data models: - SQL, - Document-oriented data, - Column-oriented data, and - Vector data.	8	Understand (Level 2)	PO3
CO2	Design and Develop solutions in Python3 and Java using MySQL, MongoDB, Vector databases.	18	Apply (Level 3) Analyse (Level 4)	PO4
CO3	Evaluate candidate data models, and choose those that are best suited for the problem at hand.	10	Evaluate (Level 5)	PO5

1.3.

1.4. Course Content (Syllabus)

Module 1	Duration: 12 Hr
Relational Data Model, SQL, ACID Guarantees. MySQL DBMS.	

Relations, Tables, MySQL DDL and DML, Transactions. Consistency, Isolation.	
Module 2	Duration: 13 Hr
NoSQL Data Model. BASE and CAP Theorem. MongoDB.	
Mongo Data Model - collections and documents. Python interfaces. Causal Consistency. BASE and CAP Theorem.	
Module 3	Duration: 11 Hr
Vector Databases. Vector Embedding. ChromaDB.	
Vector Embeddings and Vector Databases. Indexing Techniques for Vector Databases. Using LLM APIs and Creating Vector Embeddings.	

1.5.

1.4. References

1. Database System Concepts: Avi Silberschatz, Henry F. Korth, and S. Sudarshan. McGraw Hill, 6th Edition, 2010
2. NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence, Pramod J. Sadalage, Martin Fowler, Addison-Wesley, 2012.
3. MongoDB Documentation: <https://www.mongodb.com/docs/manual/>
4. ChromaDB Cookbook: <https://cookbook.chromadb.dev/core/document-ids/>
5. Chroma Documentation: <https://docs.trychroma.com/docs/overview/introduction>
6. Vector Search: <https://www.pinecone.io/learn/series/faiss/hnsw/>
7. Research Papers: MongoDB architecture, Vector databases.

Topic Learning Outcomes

Teaching-Learning Practice (TLP) - Blackboard, PPT, Quizzes, and Solving Problems in the Classroom.

Sln	Topic	CO
1	Relational Data Model, Logical and Physical Schema	CO1
2	Mathematical Ideas and their. Connection to Implementation - relations, tables, rows, and SQL	CO1
3	Functional Dependency and Schema Normalization	CO1
4	Atomicity, Consistency, Isolation and Durability - 1	CO1
5	Atomicity, Consistency, Isolation and Durability - 2	CO3
6	Principles of Transaction Processing	CO1
7	Incorporating Transactions in Python Programs	CO2
8	Principles of Transaction Processing in MySQL and PostgreSQL	CO3
9	Principles of Transaction Processing in MySQL and PostgreSQL	CO3
10	MySQL Data Definition Language	CO2
11	MySQL Data Manipulation Using Python	CO2
12	MySQL Queries, Locking, and Best Practices	CO2
13	NoSQL Movement - The necessity of Unstructured Documents	CO1

14	Document Model and NoSQL	CO1
15	Mongo Data Model - Collections and Documents	CO1
16	Mongo Data Model - Collections and Schema Specification	CO1
17	Mongo Data Model - Document Embedding and Performance Consideration	CO1
18	MongoDB DDL and DML Using Python	CO2
19	MongoDB DDL and DML Using Python - P2P Lending System Design	CO1
20	MongoDB DDL and DML Using Python - - P2P Lending System Implementation	CO2
21	Principles of Causal Consistency Model	CO1
22	MongoDB Replica Set, Primary and Secondaries, Sharding and Mongos service	CO1
23	MongoDB Causal Consistency Model - Read Concern and Write Concern	CO1
24	Comparison with BASE Consistency Model	CO1
25	CAP Theorem and Its implications	CO1
26	Introduction to Vector Databases	CO1
27	Embeddings and Vector Databases	CO1

28	Vector Operations in ChromaDB	CO2
29	Vector Support in MySQL and PostgreSQL	CO2
30	Architecture of ChromaDB	CO1
31	Data Definition and DML in ChromaDB	CO3
32	Indexing Techniques for Vector Databases	CO1
33	Hierarchical Navigable Small World (HNSW)	CO1
34	ChromaDB APIs	CO1
35	Using LLM APIs and Creating Vector Embeddings	CO2
36	Integrating ChromaDB with LLM APIs	CO2

